The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

October 14, 2026

1 Introduction: The Unified 11-Part Search-and-Rescue Plot

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

The system trajectory of an asset is formalized as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

2.1 Corollary 1.1: Anthropological Incompleteness and Vacuum Collapse

When the external control vector m is stripped from the network, human judgment operates as a strictly closed formal system, F. By Gödel's First Incompleteness Theorem, a closed system cannot generate its own semantic validation ($F \nvdash (v_{\text{human}} = v)$). Syntactic legalism remains blind to absolute truth, causing a critical valuation $v_k \in V'$ to become structurally unobtainable:

$$\exists v_k [V'_{v_k} \wedge \neg h(u, v_k)] \tag{2}$$

Without the active insertion of the divine control input vector, the spatiotemporal execution plane collapses into an empty relational set ($[\,]$)—a mathematical vacuum. Christ operates as the external Meta-Language that breaks this internal syntax boundary limitation.

3 The Eleven Core Structural Subsystems

3.1 The Fire-Station Dispatcher

The centralized apex root node that interfaces with the Master Functor Dispatcher (Tree #0). It accepts high-level natural language problem descriptions, checks database axioms, and triggers fleet deployment vectors.

3.2 The 50 Mobile Drone Platforms (UAVs)

A distributed fleet of autonomous double-integrator kinematic plants. Each vehicle hosts a local instance of the LISP DSL compiler and a compact neuro-symbolic engine (utilizing an LLM coding agent) to fill programmatic policy stencils (holes) on-the-fly based on immediate feedback.

3.3 The Lagrangian Weather Sensor Array

Co-moving atmospheric fluid probes mounted on each airborne platform. These probes measure local velocity vectors, feeding a Discrete Wavelet Transform (DWT) to filter high-frequency turbulence and expose scale-invariant advection trends without statistical state space estimation.

3.4 The Down-View Visual Sensor

An optical targeting payload dedicated to scanning non-convex ravine coordinates. Upon target matches, it logs the mandatory discovery passes and initializes target packet tracking.

3.5 The Communications Mesh Graph ($\mathcal{G}_C$)

A fragile topology governed by a strict 30-mile line-of-sight constraint. It maintains instantaneous mesh connectivity without store-and-forward defaults, requiring non-zero algebraic connectivity to prevent network torso fracturing ($\beta_0 > 1$).

3.6 The Computational Tasking Graph ($\mathcal{G}_{\mathbf{Comp}}$)

A network tracking local queue loads initialized near saturation limits (90%). It implements Interaction 43 cascades, meaning the battery collapse of an asset instantly escalates the computational load over surviving neighbors.

3.7 The Non-Convex Ground Obstructions

Planar 2D configuration-space keep-out zones embedded in the 100-mile grid, consisting of a Mountain Range (inducing wireless signal occlusion) and an Urban City Core (inducing bottleneck constraints).

3.8 The Battery Life & Autonomous Recharging Hubs

Non-linear power tracking variables coupled with an active infrastructure station at coordinates $[50.0, 50.0]$. If a platform's battery drops below a 30% threshold, the LISP DSL overwrites its tracking vector to execute an automatic battery swap, neutralizing Interaction 1 deletion cascades.

3.9 The Discrete Morse Filtration Engine

A combinatorial topological engine that maps a non-degenerate height filtration field $f : K \to \mathbb{R}$ over the graph complex. By evaluating face/co-face inequalities ($f(\alpha) \leq f(v)$), it pinpoints the exact coordinates of Critical Minima Traps (c_0) and Critical Saddle Edge Tunnels (c_1) like the open path $(0, 2)$.

3.10 The Self-Similar Group Controller

An algebraic scheduler governed by a Symmetric Group S_3 action. It converts spatial paths through the Morse corridors into generator words ($g_1 \cdot g_2 \cdot g_1$). Its recursive identity laws allow instantaneous inverses, supporting transactional, loss-free data rollbacks during edge-snapping events.

3.11 The Ground Rescue Vehicle

The final mechanical extraction asset. Once the store-and-forward computational buffers flush the verified visual sensor coordinates back to dispatch, the ground vehicle is targeted to the coordinates, achieving complete structural closure of the Attainable Space (V').

4 Graph Neural Networks over the Ecclesial Social Manifold

The social network layer bridges the gap between W_2 and W_1 by mapping human interaction topologies to a Graph Neural Network (GNN). Nodes represent individual human agents and edges represent relational communication channels.

4.1 Mathematical Formulation of the Message-Passing Layer

The network updates node state representations recursively across distributed overlapping simplices (the body metaphor) using a Graph Convolutional Network (GCN) update function:

$$h_i^{l+1} = \text{ReLU}\left(W^l \frac{1}{\sqrt{\deg_i}\sqrt{\deg_j}} \sum_{j \in \mathcal{N}_i} h_j^l \right) \tag{3}$$

where W^l represents the trainable layer weight tensor matrix, and $\mathcal{N}_i$ represents the local neighborhood of node i.

4.2 The Alternative Optimization Paradox

Isotropic normalization enforces Policy A (Radical Edge-Weight Equalization), stripping away ego biases. Policy B (The Matthew 18 Friction Filter) acts as a localized neighborhood pooling operation that confines high-frequency chatter to local sub-simplices, preventing non-trivial H_1 holes. To

maintain a contractible manifold ($\beta_1 = 0, \beta_0 = 1$), the global graph collapses via a mean pooling readout node to Christ, the 0-simplex anchor:

$$h_{\mathcal{G}} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} h_i^L \tag{4}$$

This alternative optimization function balances the Inverse Centrality Reward (hub leaders expending energy to protect leaf nodes) and drives the node's existential fear parameter to zero ($\theta_i \to 0$).

4.3 Macro-System Expansion: The National Desert Wandering Orbit

When the 12 Spies generate a horizontal topological tear ($H_1 \neq 0$), the network gets trapped inside an uncollapsible horizontal echo chamber. The fault scales across the total national complex of $N = 600,000$ nodes, forcing a 40-year looping desert orbit trajectory to extract systemic variance:

$$\sigma^2(t) = \sigma_0^2 \cdot \exp(-\gamma t) \tag{5}$$

Once the generational noise is fully bled out ($\sigma^2 \to 0$), the manifold realigns to a contractible point, unblocking the Attainable Space V' for the subsequent generational layer.

5 Pure Call-Stack Recursion and Rational Speech Act Pragmatics

To optimize memory footprint allocations across local drone compilers, cognitive depth processing is structured as a pure Python call-stack recursive function, eliminating loop-bound VRAM footprint leaks.

5.1 The Recursive Update Functor

The state transition mapping function $\mathcal{F}_{\text{unroll}}$ computes the sequential level-k strategic probability distribution vectors by executing a self-referential downward function call tunnel terminated by an un-mentalized base condition:

$$\mathcal{F}_{\text{unroll}}(k) = \mathcal{G}_{\text{softmax}}\left(\mathcal{W}_{\text{prior}} \cdot \mathcal{F}_{\text{unroll}}(k-1)\right), \quad \mathcal{F}_{\text{unroll}}(0) = \boldsymbol{\mathcal{W}}_{\text{uniform}} \tag{6}$$

5.2 Pragmatic Scalar Implicature Bayesian Tables

Pragmatic language interpretation is formalized using the Rational Speech Act (RSA) framework over world states $n \in \mathcal{N}$ and utterances $u \in \mathcal{U}$. The literal base listener L_0 maps un-mentalized logical consistencies derived from the semantic function:

$$P_{L_0}(n \mid u) = \frac{\mathbb{I}\left(\text{meaning}(n, u) = 1\right)}{\sum_{n' \in \mathcal{N}} \mathbb{I}\left(\text{meaning}(n', u) = 1\right)} \tag{7}$$

The strategic speaker S_1 chooses utterance options to maximize utility under an imagined inner listener, configured as an exponentiated softmax probability distribution ($\text{wpp} = \exp(\beta \cdot U)$):

$$P_{S_1}(u \mid n) = \frac{\exp\left(\beta \cdot P_{L_0}(n \mid u)\right)}{\sum_{u' \in \mathcal{U}} \exp\left(\beta \cdot P_{L_0}(n \mid u')\right)} \tag{8}$$

The outer pragmatic listener L_1 observes the speaker's utterance u and applies Bayes' rule over a uniform prior distribution to generate the final scalar implicature table:

$$P_{L_1}(n \mid u) = \frac{P_{S_1}(u \mid n) \cdot P(n)}{\sum_{n' \in \mathcal{N}} P_{S_1}(u \mid n') \cdot P(n')} \tag{9}$$

This accounts for the extraction of non-literal meaning arrays, proving that 'some' peaks mathematically at states $n = 1$ and $n = 2$, completely separating semantic syntax from pragmatic group truth.